

MSBA 265: Business Analytics Topics

Foundational Module 1: Practical Homework Assignment

Interactive EDA, Data Quality Verification, Data Dictionaries, & Outlier Pipelines

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Assignment Objective: Initialize a git-tracked workspace in Visual Studio Code, programmatically extract raw business data, audit raw data bounds for corruptions, construct a Business Data Dictionary artifact, plot feature distributions with written executive interpretations, build a production outlier pipeline script, and publish all reproducible artifacts to GitHub for Canvas submission.

Total Score: 100 Points (5 Parts @ 20 Points Each).

Git & Canvas Submission Guidelines (Crucial Delivery Notice)

Per our course syllabus, ****all code, notebooks, scripts, generated graphics, and compiled PDF reports must be tracked in Git and hosted on GitHub.**** You will submit a single, direct link to your public GitHub repository on Canvas for grading.

How to Submit on Canvas:

1. Push all local commits to your remote GitHub repository (`msba265-module1-yourname`).
2. Copy your public **GitHub Repository URL** (e.g., <https://github.com>).
3. Submit the URL on Canvas in the **Website URL** submission box.
4. Ensure your repository contains a root `README.md` with clear execution instructions so the instructor can clone your repository, run your scripts, and download your final compiled PDF report directly from Git.

Repository Structure & Deliverables Checklist

Your public GitHub repository must mirror this exact architecture for full credit:

```
msba265-module1-yourname/
|-- .gitignore                # Excludes venv/, __pycache__/, checkpoint files
|-- README.md                 # Contains CLI setup and step-by-step execution guide
|-- requirements.txt           # Python dependencies (pandas, seaborn, etc.)
|-- Module1_Homework_Report.pdf # Final compiled assignment PDF
|
|-- data/
|   |-- download_data.py      # Ingestion script
```

```

|     |-- raw_business_data.csv                                # Fetch output
|     |-- cleaned_business_data.csv                            # Outlier filter output
|
|-- notebooks/
|     |-- 01_eda_and_data_dictionary.ipynb                    # Interactive Jupyter Notebook
|
|-- src/
|     |-- clean_outliers.py                                    # Production filtering pipeline
|
|-- reports/
|-- data_dictionary.csv                                        # Exported data dictionary artifact
'-- figures/
|-- feature_distributions.png                                  # Distribution plots artifact
'-- correlation_heatmap.png                                    # Heatmap artifact

```

Assignment Tasks

Part 1: Workspace Setup, Git Ingestion, & Procurement (20 Points)

- 1.1 Initialize your VS Code workspace with a virtual environment (`venv`), initialize Git (`git init`), and configure a `.gitignore` excluding virtual environments and raw cache files.
- 1.2 Create and execute `data/download_data.py` to programmatically fetch the French Motor Claims dataset (`freMTPL2freq.csv`) from OpenML.
- 1.3 Commit your scripts and workspace setup to Git, and push to GitHub.

Part 2: Raw Boundary Audits & Business Dictionary Artifact (20 Points)

- 2.1 Execute `df.info()` and `df.describe().T` in Cell 3 of your Jupyter Notebook (`notebooks/01_eda_and_data`).
- 2.2 Audit raw minimum and maximum bounds for `DrivAge`, `Exposure`, and `BonusMalus`. Verify that zero negative age bugs or sentinel null codes exist.
- 2.3 Execute Cell 4 to build and export `reports/data_dictionary.csv`.
- 2.4 Write a 1-paragraph markdown defense explaining why `Exposure` acts as a rate offset, and why continuous `Density` and categorical `Area` cannot both be included in a linear regression model.

Part 3: Overriding Automated Rules & Correlation Visuals (20 Points)

- 3.1 Render the Pearson Correlation Heatmap in Cell 6b and export `reports/figures/correlation_heatmap.p`.
- 3.2 Write an executive defense explaining why you overrode automated skewness recommendations for `ClaimNb`, detailing the mathematical impossibility ($\log(0) = -\infty$) and domain rationale.

Part 4: Distribution Audits & Written Plot Statements (20 Points)

- 4.1 Render side-by-side box plots and KDE histograms in Cell 7. Export `reports/figures/feature_distribut`.
- 4.2 ****Full Credit Requirement:**** Write the required 3-bullet interpretation statements (Distribution Shape, Key Metrics/Whiskers, Recommended Analytics Action) below ****each**** of the 4 distribution charts and the correlation heatmap.

Part 5: Production Pipeline Scripting & Final PDF Export (20 Points)

- 5.1 Build and execute `src/clean_outliers.py` from your terminal to generate `data/cleaned_business_data.csv` using Tukey $1.5 \times \text{IQR}$ bounds.
- 5.2 Compile your final written answers and plot figures into `Module1_Homework_Report.pdf`.
- 5.3 Push all final commits to GitHub. Verify on GitHub that your repository is public, your `README.md` displays reproduction instructions, and your final PDF is downloadable.

Homework Answer Key & Solution Guide

Grading Rubric Summary (100 Points Total):

- **Canvas Submission Link Check:** Deduct 20 Points if the student submits raw local files to Canvas instead of a public GitHub Repository URL.
- **Part 1 (20 Pts):** Full credit for clean Git commit history, functional `.gitignore`, and 678,013 downloaded rows.
- **Part 2 (20 Pts):** Full credit for exported CSV and correct explanation of **Exposure** offsets and **Density/Area** collinearity.
- **Part 3 (20 Pts):** Full credit for heatmap export and valid mathematical defense ($\log(0) = -\infty$).
- **Part 4 (20 Pts):** Deduct 50% (10 Pts) if visual plots are submitted without written 3-bullet interpretation statements.
- **Part 5 (20 Pts):** Full credit for functional `src/clean_outliers.py` generating `cleaned_business_data.csv` and downloadable report PDF hosted on Git.

Solution: Data Quality Audit Findings (Part 2)

- **DrivAge Bound Verification:** Min = 18, Max = 100. Confirms zero negative age corruptions or child policyholders.
- **Exposure Duration Verification:** Min = 0.002 (≈ 1 day), Max = 1.0+. Confirms zero negative exposure durations.
- **BonusMalus Index Verification:** Min = 50, Max = 350. Confirms all records sit strictly within regulatory caps.

Solution: Sample Written Executive Statement for ClaimNb (Part 3)

“Although automated univariate diagnostics flag `ClaimNb` with an extreme skewness coefficient (5.60) and recommend a logarithmic transformation, we explicitly override this rule. Over 95% of policyholders record exact zero claims, making natural logarithms mathematically impossible ($\log(0) = -\infty$). Furthermore, `ClaimNb` represents discrete count events rather than continuous positive volume metrics. Therefore, `ClaimNb` must remain in raw count units and be modeled as a Poisson rate target using a $\log(\text{Exposure})$ offset.”

Solution: Sample Written Statements for Distribution Charts (Part 4)

1. **Density Box Plot:** Extremely right-skewed ($\text{skew} = 3.82$). Upper Tukey fence cuts off near 3,800 people/km². Dots between 5,000 and 25,000 represent valid metropolitan policyholders (e.g., Paris). Action: Apply $\log(1 + \text{Density})$ transformation prior to linear regression.
2. **DrivAge Histogram & KDE:** Symmetric, bell-shaped Gaussian curve centered at median age 44 ($Q1 = 34, Q3 = 55$). Action: Apply Standard Z-score Scaling ($\mu = 0, \sigma = 1$). No outlier trimming required.
3. **BonusMalus Box Plot:** Concentrated below 100 ($Q1 = 50, Q3 = 68$). Upper Tukey fence ends near 82; dots beyond 112 represent high-risk penalty drivers. Action: Scale continuously and create binary indicator flag `Is_Surcharge = BonusMalus > 100`.

4. **VehAge Histogram & KDE:** Moderately right-skewed ($\text{skew} = 1.15$), peaking between 0 and 5 years ($Q1 = 2, Q3 = 12$). Action: Bin into vehicle age brackets (0 – 5 yrs, 6 – 10 yrs, 11+ yrs) or apply standard min-max scaling.

WARNING!!! BEFORE SUBMISSION, YOU MUST SEND YOUR GIT TO A CLASS MATE. THEY MUST RUN YOUR CODE BY FOLLOWING YOUR INSTRUCTION (WITHOUT YOUR HELP!!!!). IF THEY RUN INTO ERRORS, YOU MUST FIX THEM BEFORE YOU SUBMIT YOUR FINAL SUBMISSION. INCLUDED IN YOUR DIRECTORY WILL BE A DOCUMENT STATEMENT FROM YOUR PEER THAT THEY HAVE FOLLOWED YOUR INSTRUCTIONS PROVIDED IN THE README FILE AND WERE ABLE TO SUCCESSFULLY REPLICATE YOUR PROJECT ON THEIR OWN COMPUTER. FAILURE TO DO SO WILL RESULT IN 10 POINTS DEDUCTED FROM SUBMITTER AND APPROVER FOR EACH FAILED ATTEMPT THE INSTRUCTOR MUST EXPERIENCE UPON FAILURE OF RUNNING BROKEN INSTRUCTIONS AND CODE.